

SAINT ANDREW'S JUNIOR COLLEGE

Preliminary Exam

MATHEMATICS

Higher 1

8863

Wednesday

2 September 2009

3 hours

Additional materials : Answer paper
Cover Sheet
MF15

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.

Answer **all** the questions. Total marks : 100

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically state otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematic steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

Section A: Pure Mathematics [40 marks]

1 Given that $\ln a = 20$, find the exact value of $\ln\left(\frac{a^2}{e^3}\right)$. [2]

2 Sketch the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$. On the same axes, sketch the graph $10y + x = 5$, showing its intercepts with the axes clearly. Hence, state the number of roots of the following equation $10\sin x + x = 5$ for which $0 \leq x \leq 2\pi$. [4]

3 The functions f , g and h are defined as:

$$f : x \mapsto x^2 + 3, \quad x < -1$$

$$g : x \mapsto \ln x, \quad x \in \mathbb{R}^+$$

$$h : x \mapsto x + a, \quad x < 1$$

Define each of the following functions in similar form, stating the range.

(i) f^{-1} [3]

(ii) gf [2]

Is fg a function? If not, explain why. [1]

Find the maximum value of a such that fh exists. [2]

4 (a) Sketch the graphs $y = x^2 - 2$ and $y = 4 - x$ on the same axes, indicating the points where the two lines intersect. Hence solve $x^2 + x - 6 > 0$. [4]

Shade the area, R enclosed by the two graphs and the positive x -axis. Find the exact value of the area of R . [4]

4(b) Find the area of the region enclosed by the graph of $y = \frac{1}{ax+2}$, the x -axis, the y -axis and the line $x = \frac{2}{a}$, where a is a positive constant. [2]

5 Anna is given part of a garden to grow a vegetable plot. She decides to use a rope of length 10 m to mark out a rectangular plot with dimension x m by y m. Show that the area of the plot, $A \text{ m}^2$, is given by $A = 5x - x^2$. Hence determine the dimensions of the rectangular plot so as to maximise the area of the plot. [8]

Anna's brother suggests that she mark out a circular plot to give her an even bigger area. If Anna wants to maximise the area of the plot, should she mark out a circular

or rectangular plot?

6 It is given that $f(x) = x^4 - 6x^2 + 8x + 2$.

- (i) Find $f'(x)$. [1]
- (ii) Use a non-calculator method to find the coordinates of all the stationary points and determine their nature. [4]
- (iii) Hence, sketch the graph $y = f(x)$, labeling the stationary points clearly. [1]
- (iv) From the graph, determine the possible values of k such that $x^4 - 6x^2 + 8x = k - 2$ has exactly two roots. [2]

Section B: Statistics [60 marks]

7 Three random events A , B and C defined in a finite sample space have the following relationships: The events A and B are independent and the events A and C are mutually exclusive. Given

$$P(A|B) = \frac{1}{5}, \quad P(C) = \frac{1}{10}, \quad P(A \cup B) = \frac{7}{15} \quad \text{and} \quad P(B \cup C) = \frac{23}{60},$$

Find, giving your answers in fractions,

- (i) $P(A \cup C)$ [2]
- (ii) $P(B)$ [2]
- (iii) $P(B|C)$ [3]

and state with a reason whether B and C are independent events. [1]

8 John suspects that there is some correlation between the circumference of the trunk of a particular species of tree and its age. To validate his suspicion, he collected the ages in years (x) and trunk circumferences in centimeters (y) of a random sample of eight trees of a particular species as shown in the table below.

x	11	13	40	21	28	34	45	51
y	24.4	32.1	23.8	52.8	78.3	79.2	102.7	121.2

- (i) Calculate the correlation coefficient. [1]

- (ii) Give a sketch of the scatter diagram for the data, as shown on your calculator. Giving a reason, identify a data pair which appears to be incorrect. [3]
- (iii) Remove the incorrect data pair, and calculate the correlation coefficient for the revised data. [1]
- (iv) Comment on your answer to parts (i) and (iii). [1]
- (v) Using an appropriate regression line, estimate the circumference of the trunk of a tree which is [4]
 - (a) 25 years old and
 - (b) 100 years old.
- (vi) Comment on the reliability of each of these estimates in part (v). [2]

- 9(a) Singapore Airlines wants to carry out a survey to investigate the popularity of their latest inflight entertainment system. 500 passengers in a long haul flight are classified according to the class of tickets they bought as follows:

Class	Number of Passengers
Economy	400
Business	80
First	20

Describe how you could take a stratified random sample of 60 passengers from the flight. State briefly one advantage and one disadvantage of this sampling method. [4]

- 9(b) An air steward claims that the average number of hours passengers spent on using the inflight entertainment system during a long haul flight is more than 5 hours. The number of hours a random sample of 75 passengers spent on the inflight entertainment system during a long haul flight is summarized by

$$\sum x = 393 \text{ and } \sum (x - 5.24)^2 = 162$$

- (i) Find unbiased estimates of the population mean and variance. [2]
- (ii) Test, at the 5% significance level, whether the air steward's claim is justified. [4]
- (iii) Explain what is meant by the phrase 'at 5% level of significance' in the context of this question. [1]
- (iv) The same sample is used to test the null hypothesis $\mu = \mu_0$ against the alternative hypothesis $\mu \neq \mu_0$ at 1% significance level, where μ denotes the population mean time spent on using the inflight entertainment system during [3]

a long haul flight of a passenger. Find the set of possible values of μ_o if the null hypothesis is rejected.

- 10 The mass of a steel figurine, in kg, X has a normal distribution with unknown mean μ and unknown variance σ^2 . Find the values of μ and σ , given that $P(X \leq 64) = P(X > 146) = 0.00203$. [4]

The mass of an aluminum figurine, in kg, Y is normally distributed with mean 35 and standard deviation 4. Find the probability that the total mass of 7 aluminum figurines exceed twice the mass of a steel figurine. [3]

In the market, aluminum is sold for \$2000 per kg and steel is sold for \$300 per kg. Find the probability that the mean cost of an aluminum figurine and a steel figurine is at most \$55000. [4]

- 11 A drug manufacturer claims that a certain drug cures a blood disease on average 80% of the time. To check the claim, an independent tester uses the drug on a random sample of n patients. He decides to accept the claim if k or more patients are cured.

Assume that the manufacturer's claim is true.

- (i) Find the probability that the claim will be accepted when 15 individuals are tested and k is set at 10. [2]

A more extensive trial is now undertaken on a random sample of 100 patients.

- (ii) Using a suitable approximation, find the largest value of k such that the probability of the claim being rejected is at most 1%. [6]

Assuming k is set at 75.

- (iii) Using a suitable approximation, estimate the probability that the claim will be rejected. [2]

- (iv) The independent tester collected 80 random samples of 100 patients. Find the probability that the mean number of samples that fail to meet the claim is greater than 6. [5]

- End of paper -